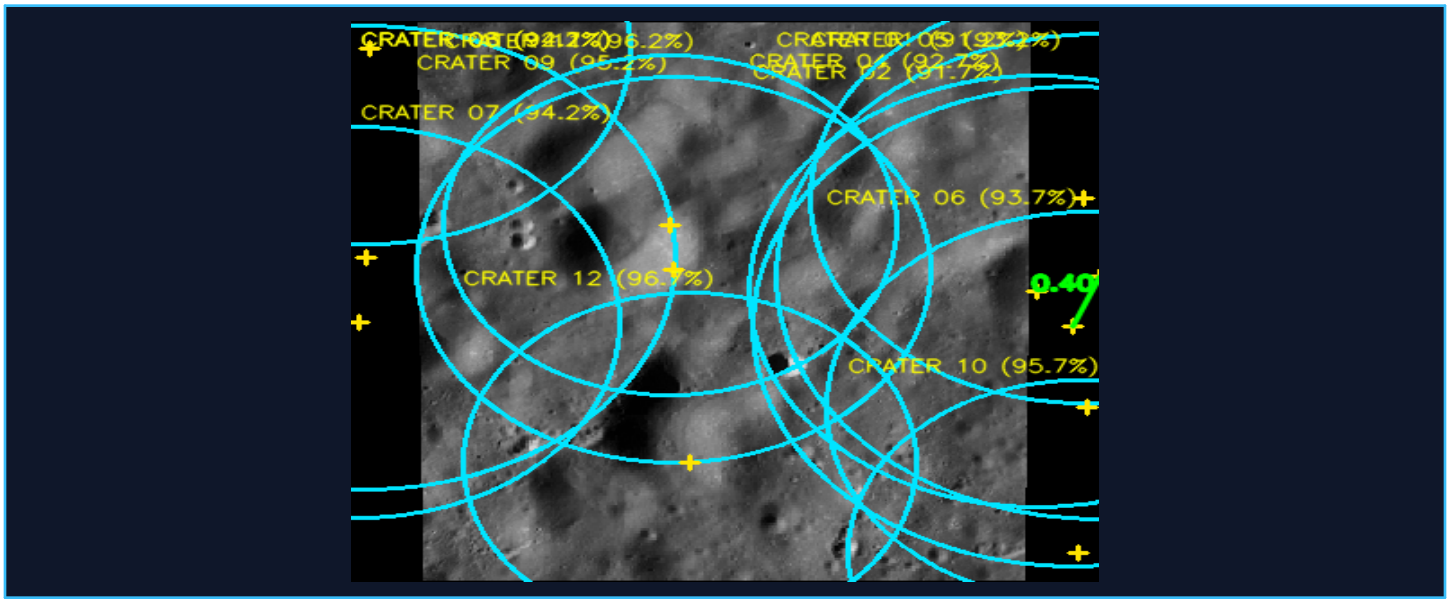
ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

Analyst / Researcher:	Dr. Alex Reynolds	**Target Body:**	**MOON**
Timestamp:	bbc5f507-694d-4fb9-	**Resolution (GSD):**	10.0 m/px
Validation Status:	**VALID_CRATER_DETECTED**	**Craters Identified:**	12
Model Pipeline:	AstroSight-CraterNet-v2.4	**Processing Time:**	367 ms

1. TOPOGRAPHICAL DETECTION VISUALIZATION



2. CRATER FEATURE COORDINATES (X, Y)

ID	Feature Name	X (px)	Y (px)	Radius (px)	Diameter (km)	Confidence
crater_01	CRATER 01	416.0	189.0	180.0	3.6 km	91.2%
crater_02	CRATER 02	401.0	226.0	178.0	3.56 km	91.7%
crater_03	CRATER 03	8.0	175.0	172.0	3.44 km	92.2%
crater_04	CRATER 04	381.0	200.0	160.0	3.2 km	92.7%
crater_05	CRATER 05	407.0	131.0	152.0	3.04 km	93.2%
crater_06	CRATER 06	409.0	286.0	145.0	2.9 km	93.7%
crater_07	CRATER 07	4.0	223.0	145.0	2.9 km	94.2%
crater_08	CRATER 08	10.0	20.0	145.0	2.9 km	94.7%
crater_09	CRATER 09	179.0	184.0	143.0	2.86 km	95.2%
crater_10	CRATER 10	404.0	394.0	128.0	2.56 km	95.7%
crater_11	CRATER 11	177.0	151.0	126.0	2.52 km	96.2%
crater_12	CRATER 12	188.0	327.0	126.0	2.52 km	96.7%

3. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

Vector	Origin Crater	Target Crater	Pixel Dist (px)	Real Distance (m)	Real Distance (km)
VEC-01	CRATER 01	CRATER 02	39.92 px	399.2 m	0.40 km
VEC-02	CRATER 01	CRATER 03	408.24 px	4082.4 m	4.08 km
VEC-03	CRATER 01	CRATER 04	36.69 px	366.9 m	0.37 km
VEC-04	CRATER 02	CRATER 03	396.30 px	3962.9 m	3.96 km
VEC-05	CRATER 02	CRATER 04	32.80 px	328.0 m	0.33 km
VEC-06	CRATER 02	CRATER 05	95.19 px	951.9 m	0.95 km
VEC-07	CRATER 03	CRATER 04	373.84 px	3738.4 m	3.74 km
VEC-08	CRATER 03	CRATER 05	401.42 px	4014.2 m	4.01 km

4. SCIENTIFIC SPECIFICATIONS & UN SDG 9

Mathematical Formulation: Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.
United Nations SDG 9 Alignment: AstroSight advances UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by establishing open AI infrastructure for planetary remote sensing, crater topology, and deep-space mission research.

Scientific Disclaimer: *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*